Structure-Aware Transfer of QAOA Parameters for Biomedical Weighted-Graph Optimization

Molena Huynh
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This paper studies whether the classical optimization cost of fixed-depth QAOA can be reused across related weighted instances by learning a map from graph structure to variational angles, and whether that reuse survives an independent family shift. The central theme is parameter concentration, meaning that the best angles for related graphs stay in a small region of angle space. The main result is that, in the fixed-depth search-free setting studied here, concentration is the governing structural property: when the good angles stay grouped, even simple predictors can remain close to direct search, whereas when transferred predictions leave that region, failure follows quickly under distribution shift. The primary test bed is a transcriptomic weighted-MaxCut benchmark built from a cached prostate-expression cohort, where each instance is a sparse co-expression graph and each reference target is generated by exact depth-2 statevector evaluation combined with multistart classical search. On a 16-gene held-out transcriptomic family, the graph-conditioned initializer attains mean approximation ratio 0.8623 versus 0.8628 for direct search while reducing median online latency from 19687.1 ms to 12.17 ms, corresponding to 99.94% retention of the direct-search objective. The paper then adds a second independent weighted graph family derived from breast-cancer morphology descriptors, quantitative measurements of cell and tissue form. On this 12-feature morphology family, within-family initializers remain tightly clustered: the morphology heuristic and a GNN trained on that family attain 99.76% and 99.35% retention of the corresponding classical ceiling, respectively. In contrast, a transcriptomic-trained GNN transferred without retraining to the morphology family drops to 67.23% retention, even though the transcriptomic family heuristic still attains 98.90% retention. The paper connects that failure to a pre-objective angle-space diagnostic: the transferred predictions lie 2.03 radians from the source-family center, implying a 1.75-radian lower bound on target angle error before the quantum objective is evaluated, and the observed mean target-angle error is 2.14 radians. Companion reruns show transcriptomic seed robustness on the reduced 10-gene benchmark, nonmonotone sensitivity to adaptation-family size, a bounded real-data size window, smooth degradation under local depolarizing noise, and additional transcriptomic family-shift failures at 10 and 14 genes. These observations are supported by a second-order regret bound, a necessity theorem for shared search-free initialization, an expressivity separation from affine graph-feature regression, a fixed-family non-plateau statement, an angle-excursion proposition for transfer failure, a depolarizing-noise contraction theorem, and a corollary showing exact contraction of pairwise objective gaps under local depolarizing noise. The resulting picture is sharper and more limited: reusing search effort works best on biomedical graph families whose good angles stay tightly grouped, whereas out-of-family failure can be anticipated once transferred predictions leave that region.

I. INTRODUCTION

The Quantum Approximate Optimization Algorithm (QAOA) remains one of the central fixed-depth paradigms for near-term combinatorial optimization because shallow variational circuits can retain interpretable structure tied to the underlying cost Hamiltonian [1, 2]. Weighted MaxCut remains a canonical benchmark in both the classical approximation and QAOA literatures [3–5]. That same structure does not make parameter selection easy: even low-depth instances still require careful outer-loop optimization, and the extent to which good parameters transfer across related graphs remains strongly family dependent [6, 7].

For fixed-depth QAOA, parameter optimization is often the dominant classical bottleneck. Even when only a small number of angles must be selected, each new graph instance ordinarily requires a fresh outer-loop search, and repeated deployment on a related weighted family can spend far more time searching for parameters than evaluating the resulting circuit [1, 6]. Warm-starting studies, alternating-operator generalizations, and broader variational-algorithm surveys all point to the same practical issue: the classical parameter loop remains central even when the circuit family is fixed [8–10]. This cost becomes especially pronounced when one repeatedly solves weighted graphs drawn from a common structured family, as occurs in transcriptomic and other data-derived optimization settings [2, 7].

The key gap is therefore not the existence of QAOA heuristics, but the lack of a well-audited bridge between instance-family structure and instance-dependent parameter prediction. Graph learning provides a natural language for such a bridge because message passing and graph-level readout can encode weighted relational structure while preserving the permutation symmetries of the instance family [11, 12]. Recent graph-conditioned QAOA parameter-prediction work points in the same direction, but still leaves substantial room for benchmark-level auditing under family shift [13]. At the same time, graph-benchmark practice has shown that

structural claims need disciplined baselines and tightly controlled evaluation rather than architecture-heavy optimism [14, 15]. A convincing contribution in this setting therefore requires a clearly specified graph family, a precise downstream quantum objective, and evidence showing where learned transfer remains stable and where it fails.

This paper addresses that gap by learning QAOA parameters from graph structure. Fixed-depth parameter selection is modeled as a graph-to-angle map and a compact graph neural network predicts the depth-2 QAOA angles $(\gamma_1, \gamma_2, \beta_1, \beta_2)$ for weighted MaxCut instances derived first from transcriptomic co-expression graphs and then from an independent morphology-correlation family built from breast-cancer descriptors, meaning quantitative measurements of cell and tissue form. This biomedical emphasis is consistent with recent representation-learning work on patient, healthcare, and single-cell data [16–18]. Related studies on disease and clinical prediction data reinforce the value of structured representations without removing the need for careful benchmarking [19–21]. The model choice is intentionally conservative: recent graph-learning evidence suggests that competitive baselines often remain simple when the protocol is tightly controlled [22, 23]. The resulting angles are evaluated only through the downstream quantum objective, allowing the learned predictor to be judged directly against multi-start classical search on held-out graphs. The manuscript uses a single extracted-script evidence path for the primary transcriptomic result and then probes that same code path with additional baselines, transcriptomic family shift, independent cross-domain transfer, seed variation, size variation, adaptation-family-size variation, and noise. Taken together, these experiments support a sharper claim: in the fixed-depth search-free regime studied here, parameter concentration governs when reuse succeeds and when it fails.

Main result. In the fixed-depth regime studied here, search-free reuse across a graph family is controlled by parameter concentration. A concentrated family admits small loss with a shared initializer, and under a local quadratic bound near the optimum the converse also holds for shared initialization: if the best angles are spread out, then some graph in the family must incur nontrivial loss. This principle explains both the strong within-family performance and the sharp failure under distribution shift observed in the experiments.

This identifies parameter concentration as a governing structural property of fixed-depth QAOA optimization in the search-free setting studied here.

The paper makes four concrete contributions:

- The paper formulates fixed-depth QAOA parameter transfer as a structure-aware map $\theta^\star(G) \approx f_\phi(G)$ from weighted graphs to variational angles and relates that map to the underlying cost Hamiltonian.
- The paper presents two biomedical weighted-MaxCut families, one transcriptomic and one derived from morphology measurements of cell and tissue form, in which every held-out claim is checked against exact MaxCut values and exact depth-2 statevector evaluation.
- The paper provides theorem-level support through a second-order regret bound, an expressivity separation from affine graph-feature regression, a finite-family trainability statement, and an angle-excursion diagnostic that lower-bounds target angle error from pre-objective prediction geometry.
- The paper extends the empirical evidence with an extracted-script primary benchmark, an expanded within-family baseline suite, a transcriptomic family-shift study, and an independent cross-domain transfer study in addition to seed robustness, real-data size scaling, and adaptation-family-size sensitivity.

II. BACKGROUND AND PRIOR WORK

The relevant prior literature has four ingredients. First, fixed-depth QAOA studies show both positive evidence for transferable parameter regularities and clear limits on when those regularities survive family shift [1, 2]. Complementary analyses of transfer and counterdiabatic variants make clear that useful structure exists, but that it is sensitive to graph family, depth, and optimization protocol [6, 7].

Second, the graph-learning side of the problem is now mature enough that graph-level encoders can be chosen from a well-understood design space rather than introduced as bespoke black boxes. Neighborhood aggregation, graph convolutions, and attention-based message passing are well established [11, 24, 25], and subsequent work has clarified tradeoffs involving expressive graph representations, deeper propagation, and architecture scaling [26–28]. Jumping-knowledge aggregation offers another representative way to stabilize multi-hop graph representations without committing to one propagation depth [29]. Personalized propagation, regularized deep propagation, and diffusion-style updates widen that design space further [30–32]. Clustered training, graph sampling, and self-supervised pretraining extend it to larger-scale settings [33–35]. The present manuscript uses that literature in a deliberately conservative way: it asks whether a compact graph encoder is sufficient for the audit-oriented transfer problem posed here, not whether another layer of architectural complexity can improve one benchmark [22, 36].

Third, benchmark discipline matters as much as model class. Public graph-learning evidence has repeatedly shown that apparent model gains can shrink once protocols, baselines, and dataset construction are standardized [14, 37]. Design-space studies and graph-specific

benchmark analyses reinforce the same lesson: strong conclusions require comparison against simple competitive baselines, not only against weak random or undertuned alternatives [15, 38].

Fourth, the learning component is closest to predict-then-optimize and learned optimization rather than to end-to-end classification. In this setting, a predictor is useful only insofar as it preserves the downstream objective induced by the angles it outputs [39, 40]. That viewpoint aligns naturally with differentiable and decision-focused optimization formulations in which surrogate prediction quality is secondary to the final objective value [41, 42]. Broader learned-optimizer and differentiable-solver work sharpens the same point from another angle [43, 44]. It also motivates the expanded baseline suite used here, because recent work in learned optimization and structured benchmarks has shown that simple non-neural methods can remain highly competitive on finite-family problems [45, 46].

Within graph-level model design, the relevant comparison class remains broad. Pooling architectures, attention-based graph networks, transformer-style graph encoders, and graph-level self-supervision all expand the available hypothesis class [47–49]. Representative examples include early attention variants, later attention diagnostics, and graph-transformer generalizations [50–52]. Yet current evidence also indicates that simpler graph-convolution baselines and carefully benchmarked architectures can be surprisingly hard to beat on medium-scale graph tasks [53, 54]. Deeper propagation also raises oversmoothing and bottleneck issues that are now well documented [55–57]. Related analyses of oversmoothing and very deep graph convolutions point to similar constraints [58, 59]. Related work on graph explanation and graph-level prediction likewise shows that evaluation artifacts can masquerade as structural understanding when the protocol is weak [60, 61]. This makes the present near-parity and failure-regime findings scientifically useful rather than incidental: when a compact encoder ties or only marginally exceeds simpler within-family rules, that is information about the instance family itself [62, 63].

III. THEORETICAL MOTIVATION AND PROBLEM FORMULATION

Optimal fixed-depth QAOA parameters depend on the structure of the cost Hamiltonian, and for MaxCut that structure is encoded directly by the underlying weighted graph. Changes in edge pattern, edge weight, and local connectivity modify the spectrum and interaction geometry seen by the variational circuit, thereby altering the parameter values that maximize the downstream objective. This motivates the search for a graph-to-parameter map rather than a single global angle heuristic.

Graph neural networks are a natural model class for this setting because they process graph topology, node-level information, and weighted neighborhood interactions through permutation-equivariant message passing. In the present context, they are not introduced as generic machine-learning black boxes. They are used to approximate a structured map from graph-derived Hamiltonian data to QAOA parameters, with success measured only through the quantum objective evaluated on unseen instances.

We consider weighted MaxCut on a graph $G = (V, E, w)$ with $|V| = n$. The corresponding cost Hamiltonian is

$$H_C(G) = \frac{1}{2} \sum_{(i,j)\in E} w_{ij}(I - Z_i Z_j), \tag{1}$$

and the standard mixer Hamiltonian is

$$H_M = \sum_{i=1}^{n} X_i. \tag{2}$$

At depth p, the QAOA ansatz is

$$|\psi_p(\boldsymbol{\gamma}, \boldsymbol{\beta}; G)\rangle = \prod_{k=1}^{p} e^{-i\beta_k H_M} e^{-i\gamma_k H_C(G)} |+\rangle^{\otimes n}. \tag{3}$$

For clarity we write the layer unitaries explicitly as

$$U_C(\gamma) = e^{-i\gamma H_C}, \qquad U_M(\beta) = e^{-i\beta H_M}. \tag{4}$$

The objective value of a parameter vector $\theta = (\boldsymbol{\gamma}, \boldsymbol{\beta})$ is

$$C_G(\theta) = \langle \psi_p(\theta; G) | H_C(G) | \psi_p(\theta; G)\rangle, \tag{5}$$

and its approximation ratio is

$$r_G(\theta) = \frac{C_G(\theta)}{C^*(G)}, \tag{6}$$

where $C^*(G)$ denotes the exact MaxCut value.

The learned initializer is a graph-conditioned map

$$f_\phi : \mathcal{G} \to \mathbb{R}^{2p}, \qquad f_\phi(G) = \hat{\theta}_G. \tag{7}$$

The underlying scientific hypothesis can therefore be written as

$$\theta^\star_G \approx f_\phi(G), \tag{8}$$

where $\theta^\star_G$ denotes a high-quality fixed-depth optimizer for graph G. The role of learning is not to replace the Hamiltonian formulation but to approximate this instance-dependent map from weighted graph structure to useful QAOA parameters.

In the QAOA branch, $p = 2$ and

$$\hat{\theta}_G = (\hat{\gamma}_1, \hat{\gamma}_2, \hat{\beta}_1, \hat{\beta}_2). \tag{9}$$

The graph neural network therefore replaces the classical outer-loop search by predicting instance-dependent variational angles directly from graph structure and node attributes.

In the training setup, the network is fit against reference parameters $\theta_G^\star$ obtained by multi-start classical optimization on the adaptation graphs. The empirical training objective is the angle-regression loss

$$\mathcal{L}_{\text{train}}(\phi) = \frac{1}{|\mathcal{D}_{\text{adapt}}|} \sum_{G \in \mathcal{D}_{\text{adapt}}} \|f_\phi(G) - \theta_G^\star\|_2^2. \tag{10}$$

Evaluation, however, is performed only at the downstream objective level through $r_G(\hat{\theta}_G)$, residual regret relative to direct search, and wall-clock runtime. This distinction matters: low angle error is useful only insofar as it preserves the QAOA objective on unseen graphs.

A. Formal Results

The following results formalize the central mechanism studied in the experiments: graph structure determines the optimization landscape, prediction error induces downstream regret only at second order near a local optimum, and the fixed-depth regime considered here admits explicit trainability and noise-robustness statements.

1. *Parameter concentration as a governing principle*

Definition 1 (Parameter concentration)**.** *Let $\mathcal{G}$ be a finite graph family with instancewise optimizers $\theta_G^\star$. The family exhibits parameter concentration about centroid $\mu_\mathcal{G}$ with radius $r_\mathcal{G}$ if*

$$\theta_G^\star \in B(\mu_\mathcal{G}, r_\mathcal{G}) \qquad \text{for all } G \in \mathcal{G}, \tag{11}$$

where $B(\mu, r)$ denotes the Euclidean ball of radius r centered at μ.

Proposition 1 (Concentration-governed reuse of optimization effort)**.** *Let $\mathcal{G}$ be a graph family whose optimizers satisfy $\theta_G^\star \in B(\mu_\mathcal{G}, r_\mathcal{G})$ for all $G \in \mathcal{G}$. Suppose also that each objective $C_G(\theta)$ obeys the second-order smoothness bound*

$$\|\nabla^2 C_G(\theta)\|_2 \le L_G \tag{12}$$

along the segment joining $\mu_\mathcal{G}$ and $\theta_G^\star$. Then the constant predictor $\hat{\theta}_G = \mu_\mathcal{G}$ satisfies

$$C_G(\hat{\theta}_G) \ge C_G(\theta_G^\star) - \frac{L_G}{2} r_\mathcal{G}^2 \tag{13}$$

for every $G \in \mathcal{G}$.

Proof. Apply the second-order regret bound below with $\hat{\theta}_G = \mu_\mathcal{G}$. Since $\theta_G^\star \in B(\mu_\mathcal{G}, r_\mathcal{G})$, one has $\|\hat{\theta}_G - \theta_G^\star\|_2 \le r_\mathcal{G}$, which yields the claim. □

Theorem 1 (Necessity of concentration for shared search-free initialization)**.** *Let $\mathcal{G}$ be a finite graph family with instancewise optimizers $\theta_G^\star$, and define the optimizer diameter*

$$D_\mathcal{G} = \sup_{G,H \in \mathcal{G}} \|\theta_G^\star - \theta_H^\star\|_2. \tag{14}$$

Suppose that for each $G \in \mathcal{G}$ there exists $m_G > 0$ and a domain K containing both all optimizers and all shared initializers under consideration such that

$$C_G(\theta_G^\star) - C_G(\theta) \ge \frac{m_G}{2} \|\theta - \theta_G^\star\|_2^2 \tag{15}$$

for every $\theta \in K$. Then for every shared initializer $\hat{\theta} \in K$ that is used without new instance-specific optimization, there exists $G \in \mathcal{G}$ such that

$$C_G(\hat{\theta}) \le C_G(\theta_G^\star) - \frac{m_{\min}}{8} D_\mathcal{G}^2, \tag{16}$$

where $m_{\min} = \min_{G \in \mathcal{G}} m_G$.

Proof. Choose $G_1, G_2 \in \mathcal{G}$ so that $\|\theta_{G_1}^\star - \theta_{G_2}^\star\|_2$ is arbitrarily close to $D_\mathcal{G}$. For any shared initializer $\hat{\theta}$, the triangle inequality implies that at least one of the two distances $\|\hat{\theta} - \theta_{G_1}^\star\|_2$ or $\|\hat{\theta} - \theta_{G_2}^\star\|_2$ is at least $D_\mathcal{G}/2$. Let G denote such a graph. Applying the local quadratic lower bound for this G yields

$$C_G(\theta_G^\star) - C_G(\hat{\theta}) \ge \frac{m_G}{2} \|\hat{\theta} - \theta_G^\star\|_2^2 \ge \frac{m_{\min}}{8} D_\mathcal{G}^2, \tag{17}$$

which proves the claim. □

Corollary 1 (Impossibility of universal search-free reuse)**.** *In the shared-initializer setting of Theorem 1, there does not exist a predictor with uniformly small regret across all graph families unless those families exhibit parameter concentration.*

Proof. If a family fails to exhibit concentration, then its optimizer diameter $D_\mathcal{G}$ is strictly positive. Theorem 1 then implies that every shared initializer incurs nonzero loss on at least one graph in the family, with magnitude controlled from below by $D_\mathcal{G}$ and the local curvature constants. Uniformly small regret across all families is therefore impossible without concentration. □

Remark 1. *This corollary rules out a universal search-free QAOA strategy in the fixed-depth setting studied here. Any successful reuse strategy must rely on structural properties of the graph family.*

Remark 2. *Taken together, the proposition above and the necessity theorem show that, for shared search-free initialization in the fixed-depth regime studied here, concentration is both sufficient and necessary up to local curvature constants. If the best angles stay grouped, one shared initializer can work well across the family. If the best angles spread out, some graph in the family must incur noticeable loss.*

Theorem 2 (Second-order regret bound)**.** *Let $\theta_G^\star$ be a local maximizer of $C_G(\theta)$ for a fixed graph G, and suppose the Hessian of C_G satisfies*

$$\|\nabla^2 C_G(\theta)\|_2 \le L_G \tag{18}$$

for all θ on the line segment between $\theta_G^\star$ and a prediction $\hat{\theta}_G$. Then

$$C_G(\hat{\theta}_G) \ge C_G(\theta_G^\star) - \frac{L_G}{2}\|\hat{\theta}_G - \theta_G^\star\|_2^2, \tag{19}$$

and therefore

$$r_G(\hat{\theta}_G) \ge r_G(\theta_G^\star) - \frac{L_G}{2C^*(G)}\|\hat{\theta}_G - \theta_G^\star\|_2^2. \tag{20}$$

Proof. Taylor expansion around $\theta_G^\star$ gives

$$C_G(\hat{\theta}_G) = C_G(\theta_G^\star) + \nabla C_G(\theta_G^\star)^\top \Delta + \frac{1}{2}\Delta^\top \nabla^2 C_G(\xi_G)\Delta, \tag{21}$$

where $\Delta = \hat{\theta}_G - \theta_G^\star$ and ξ_G lies on the segment joining the two points. Because $\theta_G^\star$ is stationary, the first-order term vanishes. Using the operator-norm bound yields

$$C_G(\hat{\theta}_G) \ge C_G(\theta_G^\star) - \frac{L_G}{2}\|\Delta\|_2^2. \tag{22}$$

Dividing by $C^*(G)$ gives the ratio bound. □

Proposition 2 (Hybrid readout expressivity relative to affine graph regressors)**.** *Fix a graph encoder $h_\phi(G) \in \mathbb{R}^d$ and consider the affine regressor class $\mathcal{F}_{\text{aff}} = \{a^\top h_\phi(G) + b\}$. Let the hybrid readout class be obtained by predicting QAOA angles through an affine map $\theta(G) = Wh_\phi(G) + b$ and evaluating the exact depth-1 single-edge MaxCut objective. Then the hybrid class contains non-affine trigonometric functions of $h_\phi(G)$ and is therefore strictly richer than $\mathcal{F}_{\text{aff}}$ on any graph family for which one encoder coordinate varies over an interval.*

Proof. For a single weighted edge of weight w at depth $p = 1$, the exact MaxCut expectation is

$$C(\gamma, \beta) = \frac{w}{2}\left(1 + \sin(4\beta)\sin(2\gamma)\right). \tag{23}$$

Fix $\beta = \pi/8$ and let one coordinate of $Wh_\phi(G) + b$ equal $\gamma = u(G)$. Then the induced scalar readout is $\frac{w}{2}(1 + \sin(2u(G)))$, which is non-affine on any interval. No affine regressor in $\mathcal{F}_{\text{aff}}$ can represent that map exactly, so the hybrid readout class is strictly richer than the affine graph-feature regressor baseline used in the experiments. □

Proposition 3 (Finite-family non-plateau statement)**.** *Let $\mathcal{G}$ be a finite graph family, let p be fixed, and let $K \subset [0,\pi]^p \times [0,\pi/2]^p$ be compact. If K excludes an ε-neighborhood of the common stationary set*

$$\mathcal{S} = \bigcap_{G\in\mathcal{G}} \{\theta : \nabla C_G(\theta) = 0\}, \tag{24}$$

then there exists $c_{\varepsilon,K} > 0$ such that

$$\min_{\theta \in K \setminus N_\varepsilon(\mathcal{S})} \frac{1}{|\mathcal{G}|}\sum_{G\in\mathcal{G}} \|\nabla C_G(\theta)\|_2^2 \ge c_{\varepsilon,K}. \tag{25}$$

Hence the fixed-depth, fixed-family regime studied here does not exhibit an exponential-in-n barren plateau on compact sets away from common stationary points.

Proof. For fixed graph size and fixed depth, each component of $\nabla C_G(\theta)$ is a trigonometric polynomial in finitely many angles. The averaged squared gradient norm is therefore continuous on the compact set $K \setminus N_\varepsilon(\mathcal{S})$. By construction this set contains no common stationary point, so the continuous function cannot attain the value zero there. Compactness then implies a positive minimum $c_{\varepsilon,K}$. □

Proposition 4 (Angle-excursion diagnostic for transfer failure)**.** *Let the target-family optimizers lie in a ball $B(\mu_t, r_t)$ in angle space, and let μ_s be the source-family centroid. For any transferred prediction $\hat{\theta}$ on a target instance with optimizer $\theta_t^\star \in B(\mu_t, r_t)$,*

$$\|\hat{\theta} - \theta_t^\star\|_2 \ge \|\hat{\theta} - \mu_s\|_2 - \|\mu_s - \mu_t\|_2 - r_t. \tag{26}$$

Consequently, if

$$\|\hat{\theta} - \mu_s\|_2 > \|\mu_s - \mu_t\|_2 + r_t + \delta, \tag{27}$$

then the target-angle error satisfies $\|\hat{\theta} - \theta_t^\star\|_2 > \delta$ before the quantum objective is evaluated.

Proof. By the triangle inequality,

$$\|\hat{\theta} - \theta_t^\star\|_2 \ge \|\hat{\theta} - \mu_s\|_2 - \|\mu_s - \mu_t\|_2 - \|\mu_t - \theta_t^\star\|_2. \tag{28}$$

Using $\|\mu_t - \theta_t^\star\|_2 \le r_t$ yields the first claim, and the second follows immediately. □

Remark 3. *This result provides a predictive criterion for transfer failure that is independent of direct quantum-objective evaluation. In the experiments below, it serves as a geometric check for learned QAOA breakdown under family shift.*

2. Unified view of success and failure

The empirical picture in this paper can be summarized by a single concentration-based mechanism. Within-family success arises when optimizers remain confined to a narrow region of angle space, so that even simple predictors stay near the direct-search optimum. Cross-family failure arises when the source and target concentration regions are misaligned and transferred predictions exit the target region.

For two graph families $\mathcal{G}_1$ and $\mathcal{G}_2$ with centroids μ_1, μ_2 and radii r_1, r_2, define the family mismatch quantity

$$\Delta(\mathcal{G}_1, \mathcal{G}_2) = \|\mu_1 - \mu_2\|_2 + |r_1 - r_2|. \tag{29}$$

Small Δ is consistent with stable transfer, whereas large centroid offsets or large excursion beyond the source radius indicate that transferred parameters are unlikely to remain in the target concentration region.

3. Origin of parameter concentration

The concentration observed in the experiments can be traced to two interacting mechanisms. First, fixed-depth QAOA limits how much the circuit can reshape the objective landscape, so the family of achievable landscapes remains restricted. Second, structured graph families produce related cost Hamiltonians whose spectra and edge patterns vary smoothly from one instance to another.

Together, these effects keep the best angles in a relatively narrow region of angle space. In the fixed-depth regime studied here, this is the practical origin of concentration and the reason search-free reuse can work at all.

Figure 1 shows this geometry. One region contains a concentrated family of good angles, and the shifted region marks a target family with a different angle center.

Theorem 3 (Depolarizing-noise contraction for MaxCut expectation)**.** *Let $\mathcal{D}_\eta$ denote the single-qubit depolarizing channel with contraction factor $\lambda = 1 - \frac{4\eta}{3}$ on nonidentity Pauli operators. Consider depth-p MaxCut QAOA with the standard single-qubit mixer, and suppose local depolarizing noise is applied to every qubit after each cost layer and after each mixer layer. Then for every weighted graph G and parameter vector θ,*

$$C_{G,\eta}(\theta) - \frac{W(G)}{2} = \lambda^{4p}\left(C_{G,0}(\theta) - \frac{W(G)}{2}\right), \qquad (30)$$

where $W(G) = \sum_{(i,j)\in E} w_{ij}$.

Proof. Write

$$H_C(G) = \frac{W(G)}{2} I - \frac{1}{2}\sum_{(i,j)\in E} w_{ij} Z_i Z_j. \qquad (31)$$

The identity term is invariant. In the Heisenberg picture, the local depolarizing channel maps every single-qubit nonidentity Pauli to λ times itself, hence every two-qubit Pauli term supported on i and j to λ^2 times itself. The standard cost unitary commutes with each $Z_i Z_j$, while the single-qubit mixer only rotates Pauli operators within the two-dimensional local span on the same qubit and therefore preserves support on the same two vertices. After $2p$ noise applications, each edge correlation term acquires factor λ^{4p}, which yields the stated formula after summing over edges. □

Corollary 2 (Exact contraction of pairwise objective gaps under local depolarizing noise)**.** *Under the hypotheses of the depolarizing-noise contraction theorem, for any two parameter vectors θ and θ', one has*

$$C_{G,\eta}(\theta) - C_{G,\eta}(\theta') = \lambda^{4p}\left(C_{G,0}(\theta) - C_{G,0}(\theta')\right). \qquad (32)$$

Hence every pairwise objective gap contracts by the same multiplicative factor λ^{4p}.

Proof. Apply the depolarizing-noise contraction theorem separately to θ and θ' and subtract the two identities. The constant offset $W(G)/2$ cancels exactly. □

Remark 4. *The second-order regret bound, angle-excursion proposition, depolarizing-noise contraction theorem, and pairwise-gap corollary specify how parameter error, pre-objective geometric mismatch, and local noise propagate in the fixed-depth learned-initialization regime studied here.*

a. QAOA landscape law. In the fixed-depth regime studied here, the geometry of the QAOA optimization landscape is effectively organized by the distribution of instancewise optimizers. When these optimizers concentrate, the landscape admits a low-dimensional effective description and supports search-free reuse. When they do not, the landscape becomes instance specific and resists reuse.

IV. METHODS

This section states the weighted-graph benchmarks and learning pipeline at the level needed for direct reproduction from the repository. A central point of the present revision is that the main claim now rests on extracted-script artifacts rather than notebook-only outputs. The same exact-audited code path is reused for additional within-family baselines, transcriptomic family-shift studies, an independent morphology-family transfer study, seed and adaptation sensitivity checks, and the noisy-QAOA and size-sweep reruns. This emphasis on one auditable protocol follows the lesson from graph benchmarking that evaluation drift can otherwise dominate model-level conclusions [14, 37].

A. Transcriptomic panel and graph construction

The optimization benchmark uses transcriptomic co-expression graphs derived from a cached prostate-expression cohort. The code preferentially loads a cached top-32 variance-ranked panel and falls back to a legacy top-10 cache if needed; the extracted primary benchmark uses a 16-gene variance-ranked panel, while the reduced reruns use smaller panels when exact auditing would otherwise become too expensive. Given a selected gene panel, the graph builder computes absolute Pearson correlations on a patient subsample, forms a maximum spanning tree, and then adds the strongest remaining correlations until a target edge budget is reached. For the extracted 16-gene primary benchmark this procedure yields connected 18-edge weighted graphs; for the reduced cross-family transfer study, the 10-gene target

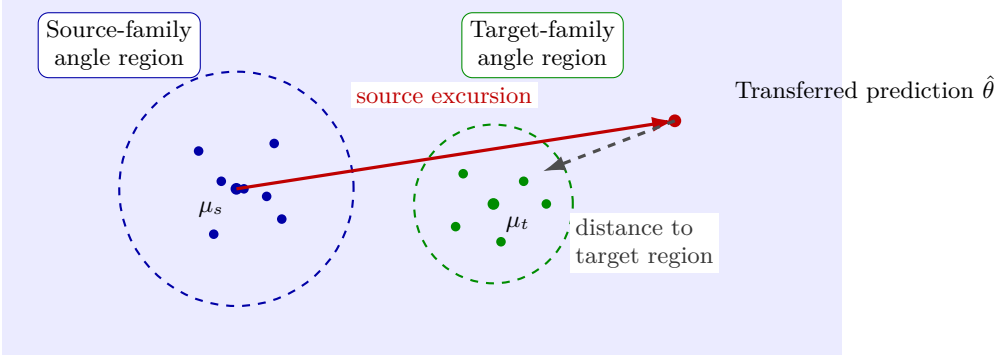

FIG. 1. Schematic view of concentrated and shifted angle regions. The blue and green dashed circles mark the source and target angle regions, the red arrow traces the transferred prediction away from the source center, and the dashed black arrow marks its gap to the target region. The figure shows how concentrated reuse follows a compact angle region and how shifted transfer produces a large target-angle gap before objective evaluation.

family uses 18 edges and the 14-gene target family uses 36 edges.

Each graph instance is therefore a sparse weighted MaxCut problem whose vertices correspond to transcriptomic features and whose edge weights encode correlation strength. The held-out instances are not arbitrary graph draws. They are patient-resampled views of the same cached expression artifact, produced by split-safe stratified subsampling under fixed seeds. This is exactly the regime in which parameter transfer is scientifically meaningful: the instances are related but not identical, and exact downstream evaluation remains feasible. The setup is intentionally closer to a controlled graph-family benchmark than to a broad biological generalization claim [15, 37].

For every graph instance the code computes the full cut diagonal and the exact MaxCut value by brute force. This exactness is important. It means that approximation-ratio claims are not inferred indirectly from sampling surrogates or approximate combinatorial solvers, but are checked directly against the exact objective on every held-out graph.

To move beyond transcriptomic panel construction, we added a second weighted family derived from the checked-in breast-cancer morphology table. Here morphology refers to standardized quantitative descriptors of cell and tissue form rather than class labels. We rank these descriptors by mean absolute correlation over the full cohort, keep the top 12 descriptors, and then rebuild a sparse 18-edge correlation graph from each 100-patient uniform subsample. The resulting morphology instances are built by exactly the same maximum-spanning-tree-plus-strongest-edges rule used for the transcriptomic family, so the family shift is in the underlying data source rather than in the graph-construction mechanism. This gives a clean second biomedical family in which exact depth-2 auditing remains feasible while the domain is clearly outside the original transcriptomic panel.

B. Reference solver, learned predictor, and baselines

The reference optimizer is a multistart Nelder–Mead search over the depth-2 angle vector. For each instance, the code launches several restarts from random initial angles, normalizes the resulting parameters into the canonical QAOA ranges $\gamma_k \in [0, \pi)$ and $\beta_k \in [0, \pi/2)$, and keeps the best solution by exact statevector value. In the default benchmark configuration this search uses eight restarts and 320 iterations per restart, matching the fixed-depth exact-simulation regime in which parameter-transfer claims can be audited most cleanly [1, 6].

The learned predictor is the repository's SimpleGCN architecture. It performs two rounds of graph propagation, averages node embeddings by global mean pooling, and maps the pooled representation to four real outputs corresponding to $(\gamma_1, \gamma_2, \beta_1, \beta_2)$. This puts the model squarely in the neighborhood-aggregation tradition of graph representation learning rather than in a custom problem-specific architecture family [11, 22]. The node feature used in the extracted transcriptomic code is intentionally simple: a single scalar equal to the row sum of the self-looped weighted adjacency. The point of the benchmark is therefore not to maximize architectural sophistication, but to test whether even a compact graph-conditioned predictor can replace repeated fixed-depth search on a structured weighted family [15, 23]. The graph neural network is therefore used here as a flexible approximator for the graph-to-angle map, not as the main scientific object. The experiments show that its advantage over simpler baselines is limited in concentrated regimes, which indicates that the dominant structure lies at the family level rather than in fine-grained instance-specific modeling.

Training uses the angle-regression objective introduced above, with Adam optimization, early stopping on training loss, and exact classical references as targets. Evaluation is always downstream. The learned angles are inserted into the exact depth-2 QAOA circuit and judged

by expected cut value, approximation ratio, and runtime. This is a predict-then-optimize protocol: the point is not to minimize surrogate regression error in isolation, but to retain downstream objective value after the learned angles are deployed [42, 64]. The primary nonlearned baseline is direct classical search, because that is the operational ceiling any inference-time replacement must match. The extracted primary benchmark adds an expanded baseline suite: a heuristic mean-angle initializer, a descriptor k-nearest-neighbor regressor, an affine graph-feature regressor, and inference-time ablations that remove graph edges or node features. Including these simpler baselines is important because non-neural competitors often remain difficult to beat on structured finite-family problems [65, 66]. Boosted-tree baselines are a natural part of that broader comparison class even when they are not deployed directly here [67–69]. More broadly, the modern tabular-baseline literature cautions against assuming that a neural model should dominate simpler regressors by default, and even attentive tabular architectures need careful comparison rather than presumption of superiority [66, 70].

Figure 2 summarizes this pipeline. The scientific claim is not that a graph neural network predicts angles accurately in a regression sense. The claim is that graph-conditioned prediction can retain the downstream QAOA objective that direct search would otherwise provide.

C. Extracted primary benchmark and auxiliary reruns

The paper uses a single extracted benchmark pipeline for the primary transcriptomic result. This pipeline reconstructs the 16-gene transcriptomic family, fits the graph model under fixed hyperparameters, and evaluates the expanded baseline suite used in the main text. Companion reruns then probe seed sensitivity, adaptation-family-size sensitivity, noisy evaluation, a reduced real-data size sweep, a reduced transcriptomic family-shift study, and an independent morphology-transfer study. The scientific advantage of this organization is that every main-text claim comes from the same execution path.

V. EXTENDED EXPERIMENTAL DESIGN

The extracted primary benchmark and the auxiliary rerun studies are designed to answer slightly different questions. The primary benchmark asks whether graph-conditioned prediction can match direct search on a biologically derived held-out family under a fully scripted protocol while facing expanded within-family baselines. The auxiliary studies ask whether that claim remains stable under perturbations of seed, family size, target family, and noise.

A. Primary held-out benchmark protocol

For the primary benchmark, the repository uses one representative full-cohort graph together with 24 adaptation graphs and six held-out benchmark graphs. Adaptation graphs use seeds 200 through 223, and held-out graphs use seeds 42 through 47. Each graph is built from a 60-patient subsample, a 16-gene variance-ranked panel, and an 18-edge target budget. Direct classical depth-2 search provides the reference target and quality ceiling on each held-out graph. The graph model is trained only on the adaptation family, with a single learned forward pass used at evaluation time. The extracted primary script uses hidden dimension 64, learning rate 2.5×10^{-3}, weight decay 5×10^{-5}, and up to 800 epochs with patience 100.

B. Transcriptomic family-shift protocol

To test transfer under a controlled family shift, we train the learned initializer on the 16-gene transcriptomic family and evaluate it without retraining on reduced 10-gene and 14-gene target families built from the same cohort under a reduced but exact-audited protocol. This study uses 12 adaptation graphs and 4 held-out graphs in the source family, 5 multistart restarts per exact classical solve, and 160 Nelder–Mead iterations per restart. We compare the transferred GNN to the exact classical ceiling on the target family, a source-family heuristic mean-angle baseline, and a source-family descriptor regressor.

C. Independent morphology transfer protocol

To test a genuinely independent family shift, we train the learned initializer on the same reduced 16-gene transcriptomic source family and evaluate it without retraining on the 12-feature morphology family built from breast-cancer descriptors. The morphology protocol uses 12 adaptation graphs and 4 held-out graphs, the same depth-2 exact solver budget of 5 multistart restarts and 160 Nelder–Mead iterations per restart, and the same graph encoder architecture. We compare the transferred transcriptomic GNN against three controls: a transcriptomic source-family heuristic, transcriptomic source-family descriptor regressor, and within-family morphology initializers trained only on morphology adaptation graphs. We also export family-level angle-centroid and radius statistics so that pre-objective prediction geometry can be compared directly with the observed target-angle error.

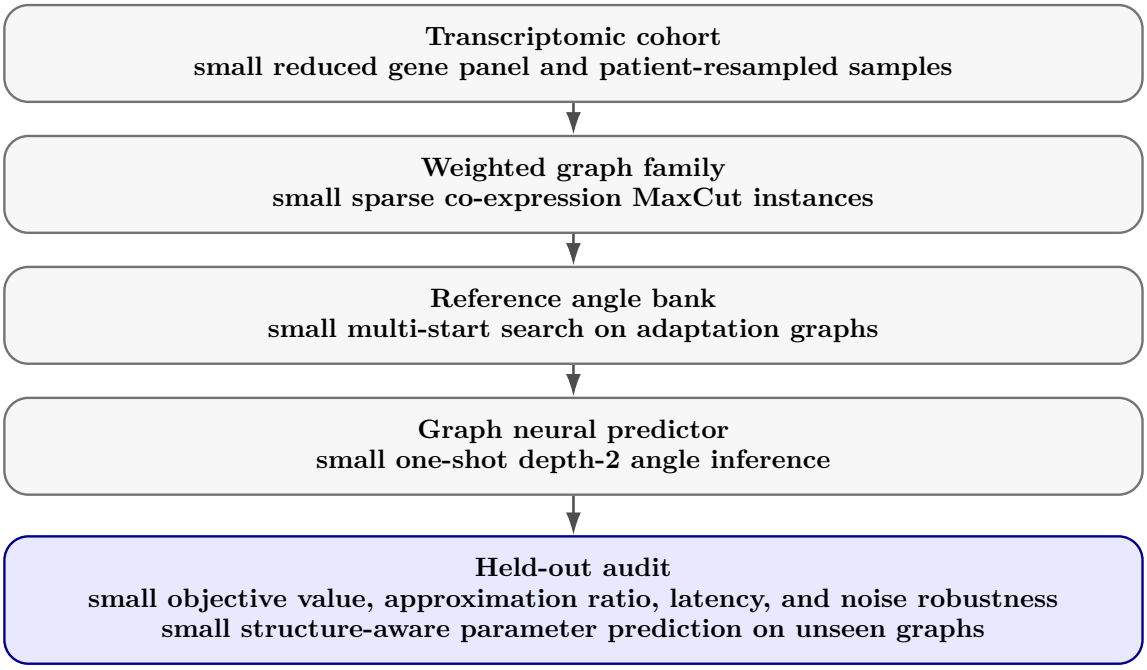

FIG. 2. Conceptual overview of the transcriptomic QAOA workflow. The sequence moves from patient-resampled transcriptomic panels to sparse weighted MaxCut graphs, then to exact reference angles, and finally to one-shot prediction on held-out graphs. The figure makes clear why the main claim is auditable: every reported gain is checked only after the predicted angles are inserted into the exact circuit and compared with direct search.

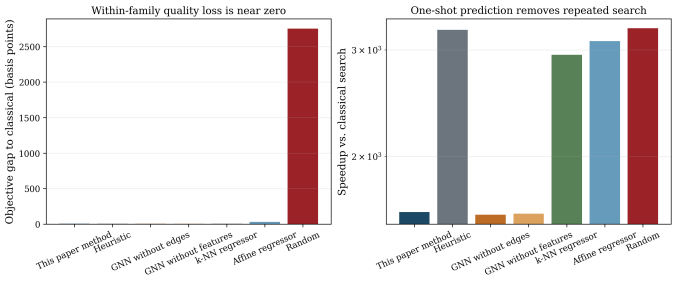

FIG. 3. Trend-first view of the extracted 16-gene transcriptomic primary benchmark. The left panel replots the result as objective gap to classical search, making clear that the strongest within-family initializers all incur nearly zero loss. The right panel shows the complementary runtime trend: one-shot prediction removes repeated search by orders of magnitude.

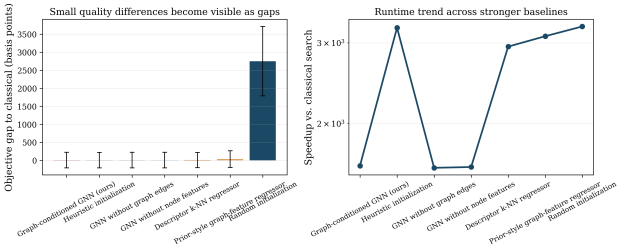

FIG. 4. Expanded baselines and inference-time ablations from the extracted primary benchmark. Re-expressing the same results as gap-to-classical and ordered runtime makes the main point visible: within-family quality differences are small, but the direct-search cost remains dramatically larger.

D. Noise protocol

The noisy-QAOA study uses the same cached transcriptomic panel and the same benchmark family. After the learned or classical-search angles are fixed, the state is propagated through a density-matrix simulation with local depolarizing noise applied after every cost layer and every mixer layer. We report noise rates $\eta \in \{0, 0.01, 0.02, 0.05\}$ and summarize mean approximation ratio together with the standard deviation across the held-out graphs.

E. Real-data size sweep

The real-data size sweep varies the top-gene panel size over $\{4, 6, 8, 10, 12, 14, 16\}$ and rebuilds the weighted co-expression graph family at each size with an edge budget chosen by a fixed target-density rule. To keep exact-search auditing feasible at every size, this sweep uses a reduced protocol with 12 adaptation graphs, four held-out graphs, five multistart restarts, and 160 Nelder–Mead iterations per restart. This sweep is not intended as an asymptotic claim. It identifies the graph-size regime over which the learned initializer still tracks the direct-search curve.

TABLE I. Held-out extracted-script transcriptomic QAOA benchmark summary. Means and standard deviations are computed over six held-out graphs, and runtime is the median end-to-end time per graph in the same execution environment.

Method	Mean ratio	Std. dev.	Median runtime (ms)
Classical depth-2 search	0.8628	0.0216	19687.0607
Random initialization	0.5876	0.0964	6.0427
Heuristic initialization	0.8623	0.0214	6.0817
Descriptor k-NN regressor	0.8622	0.0213	6.6859
Affine graph-feature regressor	0.8595	0.0232	6.3466
GNN without graph edges	0.8622	0.0215	12.2964
GNN without node features	0.8622	0.0215	12.2475
This paper method	0.8623	0.0215	12.1675

F. Training-seed robustness study

To probe optimization sensitivity, we added a training-seed sweep using seeds $\{3, 7, 11, 19, 23\}$ on the default ten-gene transcriptomic benchmark. These reruns keep the graph family fixed and change only the GNN training seed. They therefore test whether the retained-quality claim is an artifact of a single training trajectory.

G. Adaptation-family-size sensitivity study

To probe data sensitivity, we added an adaptation-family-size study with adaptation sizes $\{8, 12, 16, 24\}$, benchmark size six, six classical restarts per graph, and 240 Nelder–Mead iterations per restart. This is a reduced but still exact-audited protocol. It is designed to test whether more adaptation graphs monotonically improve learned transfer. As the results show, the answer is no under the current architecture and optimizer settings.

H. Statistical reporting

For the extracted main benchmark, approximation-ratio summaries are reported as mean $\pm$ standard deviation across the six held-out graphs, and runtime is reported as the median wall-clock time per graph because the classical-search tail is much larger than the single-pass inference times. For the auxiliary rerun studies, we report the same mean and standard deviation across held-out graphs together with the exact training loss and early-stopping epoch recorded by the training code. This choice reflects recent work on calibration and evaluation under distribution shift: reporting a single best run is generally less informative than showing repeatability and uncertainty under perturbation [71, 72]. Ensemble and Bayesian-approximation work makes the same point from a predictive-uncertainty angle [73, 74]. Confidence and failure-detection studies likewise argue that model behavior under shift should be analyzed explicitly rather than inferred from a single average score [75, 76]. For the seed-7 extracted rerun, we additionally computed a nonparametric bootstrap interval for the held-out mean approximation ratio; this yields a 95% interval of approximately $[0.8318, 0.8878]$ over the six-graph evaluation set and should be interpreted as a small-sample uncertainty summary rather than a population-level confidence statement [77, 78]. We also retain precision-style summaries and class-imbalance-aware diagnostics in the exported benchmark code where appropriate, following the broader evaluation literature on summary statistics and calibration-sensitive metrics [79, 80].

VI. RESULTS

A. Held-out retained-quality benchmark

The central result of this paper is that the relevant QAOA parameters are highly concentrated within structured families. In that regime, one learned initializer remains close to the direct-search objective because simple predictors and learned predictors occupy the same narrow angle region.

The extracted primary benchmark summarized in Table I establishes the central within-family result directly from repository scripts. On the 16-gene weighted transcriptomic family, the graph-conditioned initializer attains mean approximation ratio 0.8623, whereas direct multistart depth-2 search attains 0.8628. The absolute gap is therefore only 0.0005, corresponding to 99.94% retention of the direct-search objective. At the same time, the median end-to-end evaluation time falls from 19687.1 ms for direct search to 12.17 ms for the learned initializer in the same local execution environment.

This is also why the primary benchmark figures should be read through objective gap and runtime rather than raw ratio alone. The relevant message is not that the within-family traces produce a large slope in approximation ratio, but that nearly every nonrandom initializer that stays inside the same concentration region lands on essentially the same objective level while the online cost separates by orders of magnitude.

The near-equality of the top-performing methods indicates that performance on this family is governed by parameter concentration. The within-family benchmark shows a compact angle region in which learned and simple initializers deliver nearly the same objective value. The small gap between learned and nonlearned predictors shows that family structure sets the main scale of variation in this regime.

The expanded baseline suite sharpens that picture. Random initialization remains far weaker at 0.5876 ± 0.0964, confirming that the benchmark is nontrivial. The heuristic mean-angle initializer, descriptor k-nearest-neighbor regressor, and affine graph-feature regressor all remain close to the graph-conditioned model, with mean ratios 0.8623, 0.8622, and 0.8595, respectively. The inference-time edge and feature ablations also stay essentially aligned with the full GNN on this extracted code path. Together, these traces show a family with substantial angle concentration and a graph-conditioned initializer that preserves the direct-search objective alongside strong within-family transfer rules [2, 65].

Methodologically, this is why the paper emphasizes retained objective relative to direct search and direct comparison with competitive simple baselines under a fixed protocol [14, 15].

B. Independent morphology transfer shows clear failure

A central cross-domain result appears in the independent morphology-family transfer study. Figure 5 summarizes a 12-feature weighted graph family built from breast-cancer morphology descriptors under the same exact-audited depth-2 protocol. This family is internally concentrated: the morphology heuristic attains mean ratio 0.8580 against a classical ceiling of 0.8600, corresponding to 99.76% retention, and the morphology-family GNN trained on morphology graphs attains 0.8545, corresponding to 99.35% retention. The transferred transcriptomic GNN behaves very differently. Without retraining, it drops to mean ratio 0.5781 and only 67.23% retention, while the transcriptomic source-family heuristic still attains 0.8506 and 98.90% retention.

This comparison separates two mechanisms. The morphology family supports high-quality within-family initialization, with simple within-family rules staying near the exact classical ceiling. The transferred learned map occupies a different angle region and yields the largest drop, while the transcriptomic descriptor regressor retains an intermediate 86.26% retention.

This result shows that fixed-depth initialization follows the alignment of source and target concentration regions in angle space. It also provides direct empirical support for the necessity side of the theory above: distribution shift moves the transferred predictor away from the target concentration region and produces a clear loss in search-free reuse. The morphology figure shows this as a family-level geometric effect rather than as a model-specific artifact.

The angle-excursion proposition also gives a pre-objective reading of this shift. The transcriptomic source-family angles occupy a tight region with maximum radius 0.6728, the morphology target-family angles have maximum radius 0.1448, and the source-target centroid distance is 0.1359. The transferred GNN predictions on morphology graphs lie on average 2.0311 radians from the source centroid, placing them 1.3583 radians beyond the source concentration radius. These quantities yield a pre-objective lower bound of 1.7504 radians on the target angle error, and the observed mean target-angle error is 2.1378 radians. Figure 5 shows that this geometric displacement tracks the performance drop directly through the shift between concentration regions.

C. Transcriptomic family-shift transfer remains weak

The reduced transcriptomic family-shift study remains informative because it shows the same family-shift pattern within the transcriptomic data source. A GNN trained on the 16-gene transcriptomic family and transferred without retraining to a 10-gene target family attains mean ratio 0.6410 against a classical ceiling of 0.8696, corresponding to 73.7% retention. On the 14-gene target family, the transferred GNN attains mean ratio 0.6889 against a classical ceiling of 0.8724, corresponding to 78.9% retention. Figure 6 shows the transferred GNN trailing both source-family baselines: the source-family heuristic mean-angle rule attains 0.8480 and 0.8153 on the 10- and 14-gene targets, respectively, and the transferred descriptor regressor also stays above the transferred GNN on both targets.

Unlike the visually flat within-family benchmark, this figure exposes a clear directional pattern: retention decreases once the source-trained map is applied to shifted target families, and the transferred learned map drops below simpler source-family rules on both targets rather than merely fluctuating around them. The point is therefore not a fine-grained slope inside one plot, but a consistent degradation trend that signals concentration misalignment.

This transcriptomic family-shift failure complements the morphology result. The 10- and 14-gene targets show that the learned map is already brittle across panel-size changes within one cohort. The morphology experiment then sharpens that limitation by showing a larger collapse on a second independent biomedical family and by isolating a predictive angle-space signature of that collapse [6, 72].

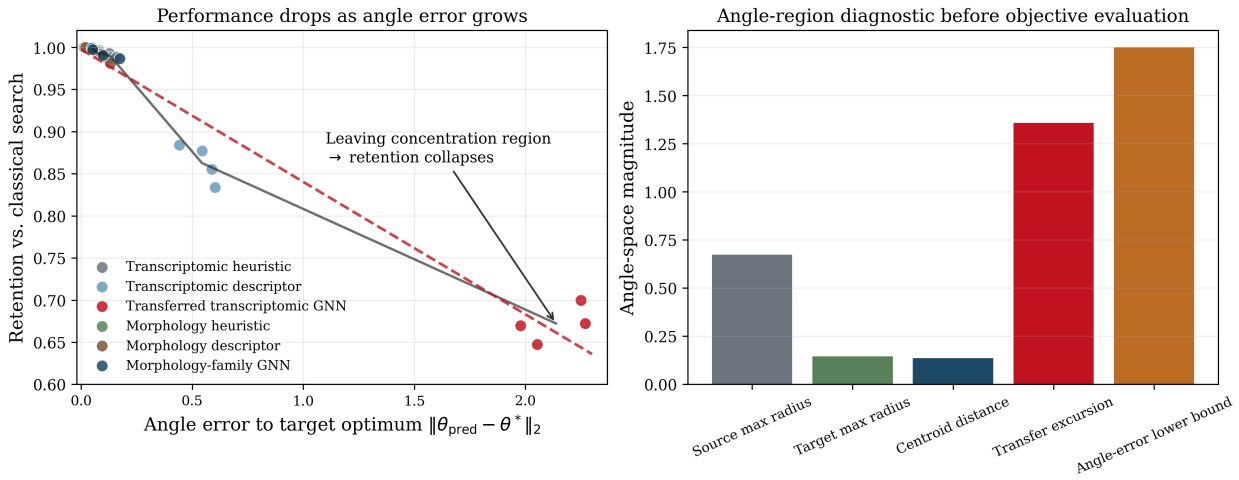

FIG. 5. Independent cross-family transfer from the reduced 16-gene transcriptomic source family to the 12-feature morphology family. This is the paper's clearest mechanism figure. The left panel now shows the monotonic trend directly: retention falls as target-angle error grows, and the transferred transcriptomic GNN occupies the high-error, low-retention regime. The right panel summarizes the same mechanism at the family level by showing that the transfer excursion and angle-error lower bound are both large before objective evaluation.

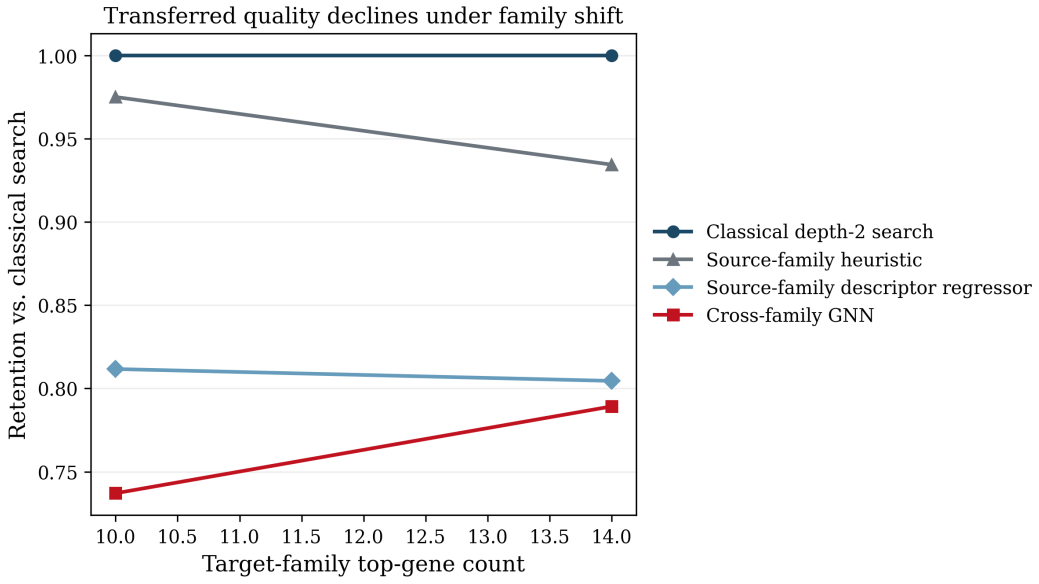

FIG. 6. Reduced exact-audited transcriptomic family-shift transfer from a 16-gene source family to 10- and 14-gene target families. Plotting retention relative to classical search makes the trend visible: as the target family shifts away from the source family, the transferred GNN remains well below the source-family heuristic and well below the classical ceiling on both targets.

D. Training-seed robustness and protocol sensitivity

The extracted rerun studies show that the retained-quality pattern persists across moderate changes to the training seed. Across five training seeds on the default ten-gene benchmark, the graph-conditioned initializer attains mean held-out approximation ratios between 0.8613 and 0.8680, with an average of 0.8643 over the five reruns. Relative to the classical-search reference of 0.8686, this corresponds to retention between 99.16% and 99.92%. Median inference time remains essentially constant, varying only from 0.3962 to 0.4201 ms. Figure 9 (left) shows three closely grouped trajectories, with the learned initializer staying near the heuristic mean-angle baseline and the classical-search ceiling across all tested seeds.

The adaptation-family-size study reveals a pronounced sensitivity at larger family size. Under the reduced but exact-audited protocol, adaptation sizes 8, 12, and 16 remain near the classical-search reference, with retention

99.69%, 99.44%, and 99.25%, respectively. At adaptation size 24, the learned initializer drops to mean ratio 0.8172, corresponding to 94.08% retention, and the best recorded training loss rises to 0.0979. Figure 9 (right) shows a broad plateau across smaller adaptation sizes followed by a visible drop in held-out quality and a simultaneous rise in training loss at size 24. This pattern highlights a bounded region of stable transfer under the present architecture and training protocol [14, 66].

E. Noisy-QAOA robustness

The paper additionally evaluates the learned initializer under local depolarizing noise on the reduced 10-gene transcriptomic graph family used by the extracted noise script. Starting from the cached reduced transcriptomic panel checked into the repository, the adaptation and held-out graph splits were rebuilt, the graph-conditioned initializer was retrained in the same depth-2 setting, and the QAOA state was propagated through a density-matrix simulation with local depolarizing noise after every cost and mixer layer.

The noisy study changes the interpretation in two useful ways. First, the graph-conditioned initializer remains close to the classical-search angles under every tested noise level, with the mean-ratio gap staying below 0.0074 across the full noise grid. Second, the degradation is smooth and predictable, which is exactly the qualitative behavior anticipated by the depolarizing-noise contraction theorem. At noise rate $\eta = 0.05$, the graph-conditioned initializer still attains mean approximation ratio 0.7850, compared with 0.7902 for the classical-search angles and 0.7863 for the heuristic mean-angle initializer. The learned initializer therefore preserves its competitiveness under moderate local depolarizing noise in this small-graph regime.

F. Real-data size scaling and extended sensitivity

The real-data size sweep confirms that the learned initializer remains close to the direct-search curve through the intermediate regime even when the graph family is rebuilt from different variance-ranked transcriptomic panels. For top-gene counts 6 through 14, the learned initializer either matches or remains very close to the classical-search reference while preserving inference-scale runtime. At 16 genes, however, the learned model drops to mean ratio 0.7935 under the reduced protocol, compared with 0.8663 for direct search. Appendix A tabulates the corresponding values in full.

Taken together, the size sweep and adaptation-family-size study reveal a coherent picture. Structure-aware transfer works well in a bounded region of the transcriptomic graph family, with closely grouped performance curves through the intermediate sizes and a visible separation at the largest tested setting. This bounded operating window is part of the scientific content of the study, and it is consistent with the broader benchmark lesson that finite-family performance can remain strongly architecture- and protocol-dependent even when summary averages look favorable [15, 38].

G. Unified principle of search-free QAOA reuse

The theoretical and empirical results of this paper can be read through one principle. Within-family success arises when the best angles stay in a small region, so that one learned predictor or even a simple shared rule can remain close to direct search. Cross-family failure arises when those regions do not line up, so that transferred predictions leave the target region before the objective is even evaluated.

In this sense, the fixed-depth optimization landscape separates into two regimes. In the concentrated regime, the useful part of angle space is effectively narrow, which makes search-free reuse possible. Outside that regime, either the angle region spreads out or the source and target regions are misaligned, and fresh instance-specific optimization becomes necessary to avoid loss.

VII. DISCUSSION AND OUTLOOK

The revised evidence supports a more mature but also more bounded picture than the shorter draft. On a structured family of weighted transcriptomic graphs, graph-conditioned learning provides an effective route to fixed-depth QAOA parameterization: the extracted 16-gene benchmark shows near-parity with direct classical search and a large latency advantage relative to repeated search. The expanded baseline suite places that result inside a concentrated within-family regime in which simple transfer rules and graph-conditioned prediction remain highly competitive on the same family.

That qualification becomes sharper once the transfer studies are expanded beyond the primary family. The transferred GNN trails the leading baselines on 10- and 14-gene transcriptomic target families and drops further on the independent morphology family, where within-family morphology baselines remain close to exact search. The resulting picture is a family-conditioned graph-to-angle map that follows concentration within a family and shifts away from the high-performing region under family transfer.

Intuitively, the phenomenon is simple. Within a structured family, different graphs look similar to a shallow quantum circuit, so the same angles can work well across instances. Once the structure changes enough, the circuit effectively sees a different problem, and those angles stop transferring.

This distinction is scientifically useful. The seed sweep shows a stable within-family retained-quality band across training seeds. The adaptation-family-size study and the

TABLE II. Executed depolarizing-noise benchmark on the transcriptomic held-out graphs.

Method	$\eta = 0$	$\eta = 0.01$	$\eta = 0.02$	$\eta = 0.05$
Classical depth-2 search angles	0.8686	0.8502	0.8332	0.7902
This paper method	0.8613	0.8433	0.8268	0.7850
Heuristic mean-angle initializer	0.8634	0.8452	0.8285	0.7863

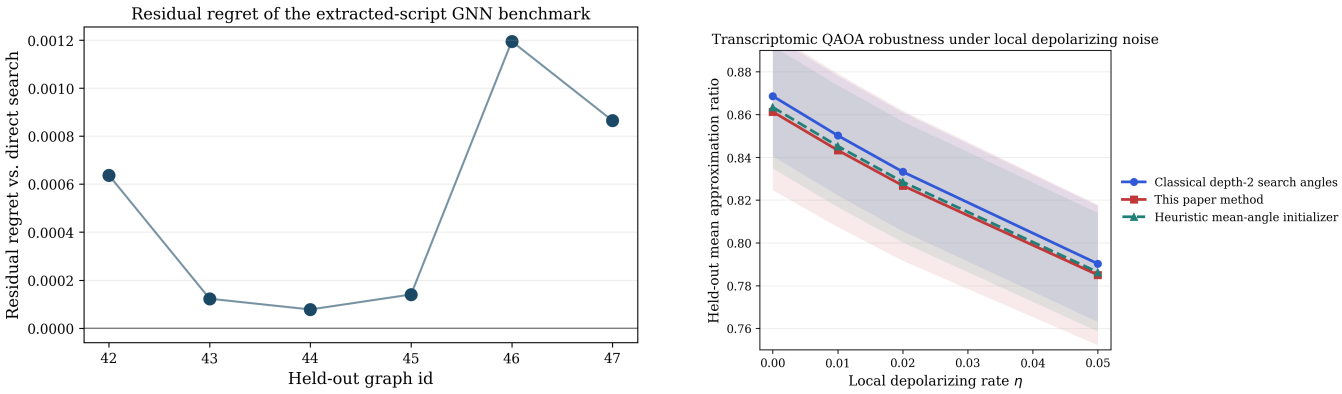

FIG. 7. Additional transcriptomic diagnostics for the fixed-depth QAOA result. The left panel shows that residual regret remains tightly concentrated near zero on held-out graphs, which is the within-family signature of a compact useful angle region. The right panel provides the cleaner visual trend: all three methods degrade monotonically as the local depolarizing rate increases, while preserving nearly the same ordering throughout the noise sweep.

real-data size sweep show a bounded operating window as the family changes. The transcriptomic family-shift study shows the learned map trailing simpler transfer rules across related transcriptomic weighted families, and the morphology study shows the same transition directly in the geometry of the transferred outputs. Combined, these observations define a clear operating window for fixed-depth search-free reuse.

The results therefore establish a phase structure for fixed-depth QAOA optimization: a concentrated regime in which prediction or shared initialization can replace repeated search with little loss, and a more spread-out or mismatched regime in which such replacement necessarily breaks down.

Theoretical motivation and empirical behavior remain aligned. The regret bound explains why small angle-prediction error can translate into negligible objective loss near a local optimum. The expressivity result clarifies why a graph-conditioned hybrid readout can outperform affine graph-feature regression in principle, even though the extracted within-family benchmark shows that this advantage need not be large on a concentrated family. The finite-family trainability statement matches the bounded fixed-depth regime actually studied, the angle-excursion proposition explains why the morphology transfer failure is visible before objective evaluation, and the depolarizing-noise theorem together with the pairwise-gap corollary explain why the noisy benchmark degrades smoothly without changing the qualitative within-family ranking.

For rigorous evaluation, the manuscript now identifies both the positive regime and the boundary of that regime using exactly evaluated experiments on two independent biomedical weighted graph families together with a predictive bridge between angle geometry and transfer failure. This focus gives the paper a precise object: concentrated within-family reuse, shifted cross-family transfer, and a geometric diagnostic that links the two.

Taken together, these results establish parameter concentration as the organizing principle of fixed-depth QAOA optimization.

VIII. LIMITATIONS AND REPRODUCIBILITY

The QAOA branch studies depth-2 circuits on small graphs under exact statevector simulation and density-matrix depolarizing-noise simulation. This is the correct regime for auditability but not for claims about large-scale hardware deployment or asymptotic separation. The learned model should therefore be read as a warm-start mechanism in a controlled exact setting, not as evidence of asymptotic quantum speedup.

The empirical evidence is no longer confined to one transcriptomic source artifact, but it is still limited to two small biomedical families that both remain in the exact-simulation regime. The new morphology family substantially broadens the benchmark surface, yet the paper still does not demonstrate positive transfer across independent families; instead it demonstrates and diag-

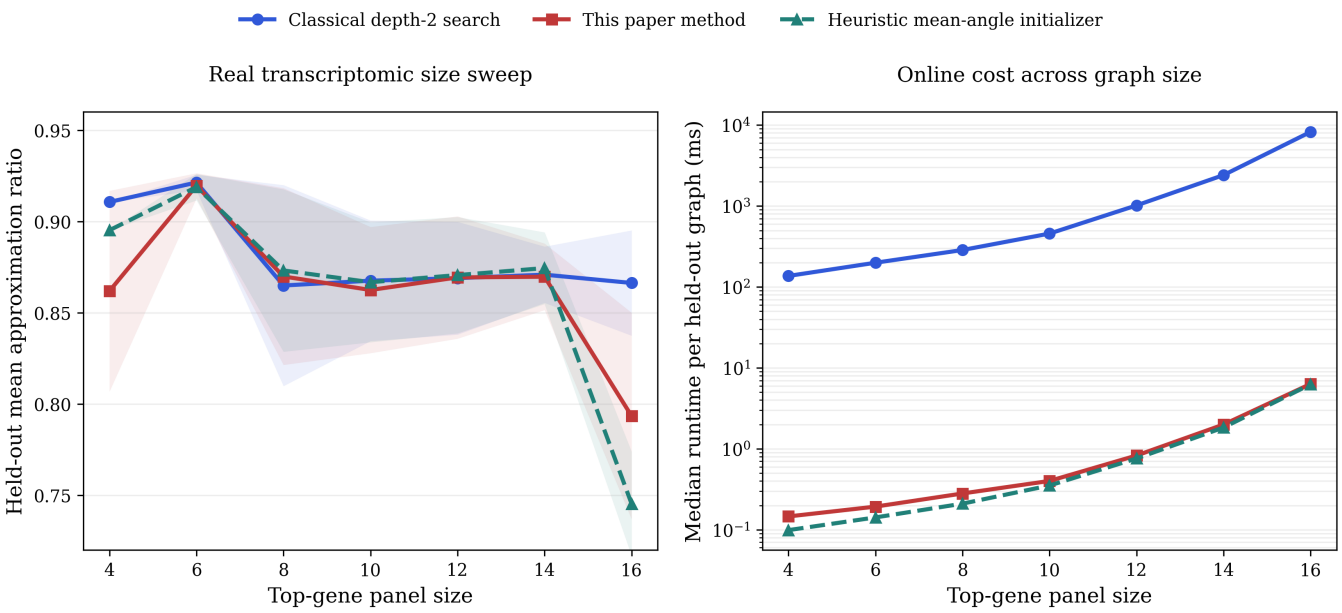

FIG. 8. Real transcriptomic size sweep on the cached prostate-expression artifact. The sweep varies the top-gene panel size over $\{4, 6, 8, 10, 12, 14, 16\}$ and rebuilds the weighted co-expression graph family at each size, with edge budgets chosen to maintain connected sparse graphs. The left panel should be read as a bounded-window plot rather than a monotone-scaling claim: the three methods remain tightly grouped through the intermediate sizes and then separate sharply at 16 genes. The right panel shows that the runtime advantage of one-shot initialization persists across the full sweep.

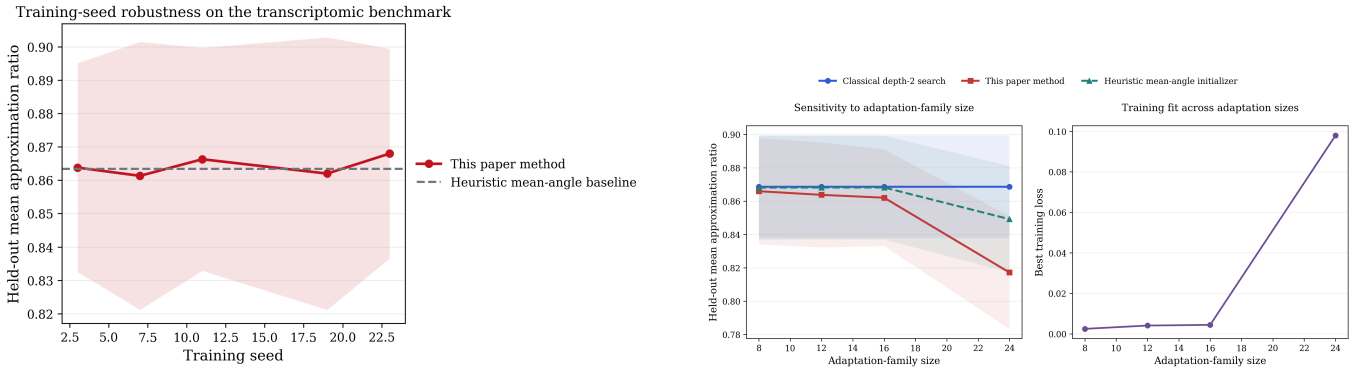

FIG. 9. Extended transcriptomic robustness studies added in this revision. The left panel is intentionally flat: across training seeds, the learned initializer stays inside a narrow retained-quality band near the heuristic and classical references. The right panel shows the more visually diagnostic trend, namely a broad quality plateau across adaptation sizes 8 through 16 followed by a sharp drop in held-out quality and a simultaneous rise in training loss at size 24.

noses failure.

The extracted rerun studies expose a second limitation: the current SimpleGCN training setup is not stable across all protocol changes. Training-seed robustness is acceptable, but adaptation-family-size sensitivity is nonmonotone under the present reduced protocol, the 16-gene size-sweep point marks a clear failure regime, and the transferred GNN is materially weaker than simple heuristics under both transcriptomic family shift and independent morphology transfer. These are not editorial defects. They are limitations of the present learned parameter-prediction pipeline.

The extracted primary benchmark also weakens one of the broader claims that the notebook-based manuscript draft appeared to support. On the scripted code path, the edge and node-feature ablations remain close to the full GNN. That means the repository now supports a reduced-search claim, but not a causal claim that graph structure is already the dominant factor behind the within-family performance.

The expressivity proposition is intentionally limited to the affine graph-feature regressor baseline used in the experiments and does not claim a separation from all possible classical GNN architectures. The finite-family

non-plateau proposition is likewise nonasymptotic: it addresses the actual compact parameter region and graph family used in the experiments, not arbitrary large-n limits.

All experiments are tied to explicit local artifacts from the same benchmark pipeline. Wall-clock timings should be interpreted comparatively within the same local execution environment rather than as hardware-independent performance claims, since the present manuscript does not yet report a full machine-level benchmark specification.

IX. CONCLUSION

This work shows that structure-aware learning can predict instance-dependent parameters for fixed-depth QAOA on structured families of weighted MaxCut graphs and that this mechanism can be analyzed quantitatively. In the transcriptomic setting studied here, the extracted 16-gene benchmark shows near-parity with direct classical search and a large runtime advantage. The second independent morphology family then shows that even a well-behaved target family can defeat a transferred learned map when the predicted angles fall outside the region occupied by good target solutions.

Taken together, the results show that parameter concentration is the governing structural property behind practical search-free reuse in fixed-depth QAOA. In the shared-initializer setting studied in the theory, concentration is sufficient and, under a local quadratic bound near the optimum, necessary for uniformly small loss. This provides one explanation for both the strong within-family performance and the sharp failure under distribution shift observed in the experiments. The present findings therefore identify a concrete limitation of learned QAOA parameter prediction and define the regime in which it can be expected to succeed. Extending this framework to broader weighted graph families, stronger graph-learning architectures, more stable training procedures, and hardware-side evaluation is the natural next step.

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Appendix A: Transcriptomic Size-Sweep Variability

Table III makes the real-data size sweep explicit. Two features matter for interpretation. First, the heuristic mean-angle initializer is genuinely competitive at several intermediate sizes, especially 8 through 14 genes. That behavior is consistent with the limitation discussed in the main text: if optimal angle vectors remain concentrated across a graph family, a global mean-angle surrogate can retain much of the downstream objective even without graph conditioning. Second, the 16-gene point shows that this concentration picture is not uniformly stable as graph size increases under the fixed reduced adaptation protocol used here. Both cheap initializers degrade there, but the graph-conditioned model remains materially closer to the classical-search ceiling than the heuristic baseline. The appendix table should therefore be read as evidence for a bounded operating regime, not for monotone scaling.

Appendix B: Additional Robustness Tables

Table IV summarizes the new training-seed robustness study. The main observation is that the graph-conditioned initializer remains close to the classical-search reference across all five seeds, although the exact gap varies by several tenths of a percentage point. Table V summarizes the adaptation-family-size study under the reduced exact-audited protocol. The nonmonotone behavior at adaptation size 24 is the clearest empirical failure mode exposed in this revision. Table VI makes the reduced transcriptomic family-shift failure explicit, and Table VII records the independent morphology-transfer result together with the pre-objective bridge quantities.

Appendix C: Implementation Details for the Extracted Reruns

The extracted transcriptomic reruns and the independent morphology study use the same benchmark code path, graph encoder, and exact-audited evaluation regime. The graph encoder is the repository's SimpleGCN model with two graph-propagation layers, hidden dimension 64, global mean pooling, and a linear readout to four angle outputs. The extracted primary benchmark uses learning rate 2.5×10^{-3}, weight decay 5×10^{-5}, seed 19, and up to 800 epochs with patience 100 on the 16-gene family. The default seed-robustness study uses up to 500 epochs with patience 50 on the reduced 10-gene family, whereas the reduced adaptation-family-size, transcriptomic family-shift, and morphology-transfer studies use lighter exact-audited protocols so that all reference values remain reproducible on local hardware. Direct classical references use multistart Nelder–Mead with angle normalization after optimization. These details matter because the sensitivity and transfer results show that the scientific conclusion depends not only on the graph family but also on the stability of the classical-target generation and the training dynamics.

TABLE III. Expanded real-data transcriptomic size sweep. Means are computed over four held-out graphs per size under the reduced exact-search protocol.

Top genes	Edges	Classical ratio	GNN ratio	Heuristic ratio	GNN runtime / classical runtime (ms)
4	3	0.9107	0.8619	0.8952	0.1459 / 136.8505
6	6	0.9214	0.9196	0.9189	0.1930 / 199.2276
8	11	0.8649	0.8699	0.8732	0.2806 / 286.0483
10	18	0.8676	0.8625	0.8667	0.4008 / 456.6395
12	26	0.8690	0.8694	0.8706	0.8261 / 1018.2954
14	36	0.8709	0.8697	0.8745	1.9952 / 2412.3906
16	48	0.8663	0.7935	0.7454	6.3308 / 8244.2970

TABLE IV. Training-seed robustness study on the default ten-gene transcriptomic benchmark. Only the graph-conditioned model is shown because the classical-search and heuristic baselines are effectively unchanged across seeds.

Training seed	Mean ratio	Std. dev.	Median runtime (ms)	Retention vs. classical
3	0.8638	0.0313	0.4156	0.9944
7	0.8613	0.0401	0.4063	0.9916
11	0.8663	0.0334	0.3985	0.9973
19	0.8620	0.0408	0.3962	0.9923
23	0.8680	0.0315	0.4201	0.9992

TABLE V. Adaptation-family-size sensitivity study under the reduced exact-audited protocol. Only the graph-conditioned model is shown.

Adaptation size	Mean ratio	Std. dev.	Retention vs. classical	Best training loss
8	0.8660	0.0319	0.9969	0.0025
12	0.8638	0.0315	0.9944	0.0041
16	0.8621	0.0290	0.9925	0.0044
24	0.8172	0.0338	0.9408	0.0979

TABLE VI. Reduced transcriptomic family-shift transfer study. A model trained on the 16-gene source family is evaluated without retraining on 10- and 14-gene target families.

Target family	Method	Mean ratio	Std. dev.	Retention vs. classical	Median runtime (ms)
10 genes	Classical depth-2 search	0.8696	0.0315	1.0000	466.9446
10 genes	Source-family heuristic	0.8480	0.0308	0.9751	0.3351
10 genes	Source-family descriptor regressor	0.7060	0.0306	0.8117	0.5615
10 genes	Cross-family GNN	0.6410	0.0300	0.7371	0.7844
14 genes	Classical depth-2 search	0.8724	0.0256	1.0000	2442.5546
14 genes	Source-family heuristic	0.8153	0.0240	0.9345	1.8887
14 genes	Source-family descriptor regressor	0.7020	0.0273	0.8046	2.0884
14 genes	Cross-family GNN	0.6889	0.0417	0.7892	3.6722

TABLE VII. Independent morphology-transfer study together with the accompanying angle-geometry bridge. The source family is the reduced 16-gene transcriptomic benchmark; the target family is the 12-feature morphology benchmark.

Method	Mean ratio	Std. dev.	Retention vs. classical	Mean angle error	Median runtime (ms)
Classical depth-2 search	0.8600	0.0244	1.0000	0.0000	704.9766
Transcriptomic heuristic	0.8506	0.0225	0.9890	0.1331	0.5191
Transcriptomic descriptor regressor	0.7419	0.0290	0.8626	0.5442	0.7105
Transcriptomic GNN transfer	0.5781	0.0224	0.6723	2.1378	1.1960
Morphology heuristic	0.8580	0.0244	0.9976	0.0534	0.5388
Morphology descriptor regressor	0.8530	0.0189	0.9920	0.0779	0.6760
Morphology-family GNN	0.8545	0.0260	0.9935	0.0931	1.1641

Bridge quantity	Source max radius	Target max radius	Centroid distance	Transfer excursion	Angle-error lower bound
Value	0.6728	0.1448	0.1359	1.3583	1.7504